

MINGQING XU

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EDUCATION

The Hong Kong University of Science and Technology (Guangzhou) *Sep 2023 - Oct 2025*
MPhil in Computational Media and Arts *Guangzhou*

- Advisor: Prof. [Mingming Fan](#)
- Thesis: *A Projection Notification System for Improving Students' Posture Unobtrusively*

South China University of Technology *Sep 2019 - Jul 2023*
Bachelor of Engineering in Industrial Design *Guangzhou*

- Advisor: Prof. Dan Liao (Graduation thesis)

RESEARCH INTEREST

I am interested in research at the intersection of technology and human well-being, with a particular emphasis on how emerging technologies such as artificial intelligence (AI), mixed reality (MR) can support both physical health and mental health.

RESEARCH EXPERIENCE

Augmenting People by Empowering X (APEX) Lab, HKUST(GZ), supervised by Prof. Mingming Fan

An ambient projection system for posture correction during desk-based learning

Sep 2024 - Sep 2025

- **Research:** This study investigated how to design an ambient projection notification system that delivers non-intrusive posture correction cues during desk-based learning. I first examined users' perceptions of fundamental projection notification elements, then derived a design space for desk-based projection notifications through a co-design workshop. Building on these insights, I developed and evaluated an ambient system that supports multiple notification strategies for posture correction.
- **Related Skills:** Unity development (C# scripts, Shader Graph component), MediaPipe plugin for posture recognition, qualitative data analysis using thematic analysis, quantitative data analysis using RM-ANOVA, Wilcoxon Signed-rank test.
- **My Task:** Project leader, system implementation, user study, data analysis, draft writing
- **Status:** Accepted by CHI 2026 (First Author)

A Systematic Review of Reminiscence Technologies in HCI

Jun 2025 - Sep 2025

- **Research:** We conducted a literature review on reminiscence technologies in the HCI field following the PRISMA workflow, and derived a cue-centered framework that characterizes different design mechanisms for cue-centered reminiscence.
- **Related Skills:** Literature review data analysis, Theoretical framework conceptualization
- **My Task:** Co-author, framework visualization, data analysis
- **Status:** Accepted by CHI 2026 (Second Author)

Toward Facilitating Search in VR With the Assistance of Vision Large Language Models

Mar 2024 - Oct 2024

- **Research:** We investigated how users search for information in Virtual Reality (VR) and identified six key search needs through a literature review and formative study. We then designed six probes powered by Vision-Large Language Models (V-LLMs) to explore potential solutions. A user study on these probes led us to derive design principles for creating more intuitive and efficient search interfaces in VR.
- **Related Skills:** VR scene construction, qualitative data analysis (thematic analysis)
- **My Task:** Co-author, probe design, user study

- **Status:** Accepted by VRST 2024 (co-author)

South China University of Technology, supervised by Prof. Dan Liao

VR experience designed for mental health

Jun 2022 - 2023 Jun

- **Research:** We designed and developed interactive VR scenes and experiences based on psychological theories, such as positive psychology and bereavement counseling techniques.
- **Related Skills:** Unity development (C# scripts, XR Interaction Toolkit)
- **My Task:** Interactive VR scene development, User study
- **Status:** One subproject in which I participated was accepted at AIID 2022 (co-author), while another subproject that I led was expanded into my undergraduate thesis.

PUBLICATIONS AND MANUSCRIPTS

Mingqing Xu, Zhongyue Zhang, Zhiyuan Xia, Chao Liu, and Mingming Fan (2026). PosProjector: An Ambient Projection Notification System for Posture Correction During Desk-Based Study. In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26). [paper](#)

Zhongyue Zhang, **Mingqing Xu**, Mengyang Wang, Lina Xu, Xu Zhang, and Mingming Fan(2026). From Memory to Meaning: A Systematic Review of Reminiscence Technologies in HCI. In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26). [paper](#)

Chao Liu, Chi San (Clarence) Cheung, **Mingqing Xu**, Zhongyue Zhang, Mingyang Su, and Mingming Fan (2024). Toward Facilitating Search in VR With the Assistance of Vision Large Language Models. In Proceedings of the 30th ACM Symposium on Virtual Reality Software and Technology (VRST '24). [paper](#)

Dan Liao, Yi Xia, **Mingqing Xu**, and Mengqi Ding (2023). A grief farewell scene design for children based on virtual reality. In International Conference on Artificial Intelligence and Industrial Design (AIID 2022). [paper](#)

SKILLS

Development and software

Unity – Experienced in developing camera projection system and VR applications using C# and designing custom interactive visual effects with the Shader Graph component.

P5.js & Processing – Create interactive artworks through creative coding.

Blender – 3D modeling, rendering, animation production and procedural content generation through Geometry Nodes.

UI/UX Design – Experienced in user interface design using Figma, and building interactive web using HTML, CSS, and JavaScript.

Creativity tools - Video editing and graphic design using Adobe Kit(PR, AI, PS...)

Qualitative data analysis - NVivo

Industrial design tools - Rhino, KeyShot

Languages

English (Fluent, TOEFL iBT: 105 (R27, L29, S23, W26)), Mandarin (Native)

AWARDS AND SCHOLARSHIPS

Postgraduate Studentship (~1,400 USD per month) *2023–2025 Academic Year*

Merit Student, School of Design *2020–2021 Academic Year*

Second-Class Scholarship, Third-Class Scholarship *2020–2021, 2021–2022 Academic Year*

Hong Kong Digital Art and Design Competition, Third Prize *2023*